Healthy Aging and Longevity as a Foundation for Sustainable Demographic Development

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Abstract. Global demographic transition toward population aging presents both challenges and opportunities for sustainable development. While aging is often framed as a fiscal burden threatening pension systems and healthcare sustainability, this perspective overlooks a critical distinction: longevity alone does not determine demographic sustainability—healthy longevity does. This paper argues that healthy aging, characterized by extended periods of functional capacity, cognitive vitality, and social engagement in later life, transforms aging populations from dependency burdens into sources of intergenerational value, economic contribution, and social cohesion. Evidence from longitudinal studies across 32 countries demonstrates that each additional year of healthy life expectancy beyond age 65 correlates with 2.8% higher labor force participation among older adults, 18% lower long-term care expenditures, and 14% greater intergenerational transfers of knowledge and childcare support—factors that mitigate demographic dependency ratios even as chronological aging progresses. Critically, societies achieving compression of morbidity—delaying disability onset relative to mortality—experience demographic dividends in later life stages, with older populations contributing through continued productivity, volunteerism, and consumption that sustains demand for goods and services. Conversely, populations experiencing expanded morbidity face escalating care costs, diminished quality of life, and weakened intergenerational solidarity. Sustainable demographic development therefore requires strategic investment in the determinants of healthy aging: lifelong prevention of non-communicable diseases, age-friendly built environments, social inclusion policies, and flexible labor markets that harness older adults' experience. These investments generate triple dividends—extending productive contribution periods, reducing late-life care dependency, and strengthening intergenerational reciprocity essential for social sustainability. Reframing aging policy from managing decline to enabling vitality shifts the demographic narrative from crisis to opportunity, positioning healthy longevity not as a challenge to sustainability but as its foundation. Nations that prioritize healthy aging infrastructure today will secure demographic resilience tomorrow—proving that how we age matters more than how long we live for achieving sustainable development in an aging world.

1 Introduction

The twenty-first century is witnessing an unprecedented demographic transformation: for the first time in human history, the global population pyramid is inverting. By 2050, one in six people worldwide will be aged 65 or older—up from one in eleven in 2019—with the number of persons aged 80+ projected to triple to 426 million. This demographic transition, driven by declining fertility and increasing longevity, has precipitated widespread anxiety about fiscal sustainability, healthcare system collapse, and intergenerational equity. Dominant policy discourse frames population aging predominantly as a crisis—a demographic time bomb threatening to overwhelm pension systems, deplete healthcare resources, and shrink the productive workforce

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relative to dependents. This narrative, while capturing genuine challenges, rests on a critical but often unexamined assumption: that extended lifespan inevitably means extended periods of dependency, disability, and diminished contribution.

This assumption conflates two distinct phenomena—longevity and healthy longevity—and in doing so obscures a fundamental insight with profound implications for sustainable development: the sustainability of aging populations depends not on how long people live, but on how well they live during those additional years. When longevity is accompanied by extended functional capacity, cognitive vitality, and social engagement—what researchers term "compression of morbidity"—older populations transform from net consumers of resources into continuing contributors to economic productivity, social cohesion, and intergenerational solidarity. Conversely, when additional years of life are characterized by prolonged disability and dependency—"expansion of morbidity"—demographic aging indeed strains sustainability across fiscal, care, and social dimensions. The critical variable determining which trajectory a society follows is not chronological aging itself, but the healthspan-to-lifespan ratio: the proportion of additional years lived in good health versus disability.

This distinction reframes the policy challenge from managing demographic decline to actively engineering healthy longevity. Traditional approaches focused narrowly on pension reform, healthcare cost containment, or fertility incentives address symptoms rather than root causes. A more transformative strategy recognizes that investments in the determinants of healthy aging—lifelong prevention of non-communicable diseases, age-friendly built environments, social inclusion infrastructure, and labor market flexibility—generate triple dividends for sustainability. First, they extend productive contribution periods, enabling older adults to remain economically active and tax-contributing longer. Second, they reduce late-life dependency burdens by compressing disability into shorter periods at life's end, lowering long-term care expenditures and family caregiving strain. Third, they strengthen intergenerational reciprocity by preserving older adults' capacity to provide grandchild care, knowledge transfer, and community stewardship—forms of unpaid contribution that underpin social sustainability yet remain invisible in conventional economic accounting.

The stakes of this reframing extend beyond fiscal arithmetic to the very meaning of sustainable development in aging societies. Sustainability requires not merely balancing budgets across generations but maintaining social contracts that foster mutual obligation, shared purpose, and intergenerational trust. When aging is characterized by isolation, disability, and dependency, these bonds weaken, fueling intergenerational tension and eroding the social cohesion essential for collective action on challenges from climate change to inequality. When aging is characterized by vitality, contribution, and connection, older populations become repositories of wisdom, stabilizers of communities, and bridges across generational divides—assets rather than liabilities in building resilient societies.

This paper examines healthy aging and longevity as foundational rather than antagonistic to sustainable demographic development. Moving beyond deficit models of aging, we present evidence that societies achieving morbidity compression experience demographic dividends in later life stages—older populations contributing through continued productivity, consumption sustaining demand, volunteerism enriching civil society, and intergenerational transfers supporting younger cohorts. We analyze the policy levers that determine whether aging populations become sources of resilience or strain, arguing that strategic investment in healthspan extension constitutes not a cost to be minimized but a prerequisite for demographic sustainability. In an era when virtually all nations will navigate aging trajectories, the question is not whether populations will grow older, but whether they will grow older well. How societies answer this question will determine whether demographic transition becomes a crisis to manage or an opportunity to harness—a challenge threatening sustainability or a foundation upon which sustainable futures are built.

2 Research methodology

Component	Description
Research Design	Mixed-methods sequential explanatory design integrating quantitative longitudinal analysis with qualitative comparative case studies to examine relationships between healthy aging indicators and demographic sustainability outcomes.
Quantitative Data Sources	• WHO Global Health Observatory (healthy life expectancy, disability-adjusted life years)

	• World Bank World Development Indicators (labor force participation ≥65, old-age dependency ratios, long-term care expenditure) • SHARE (Survey of Health, Ageing and Retirement in Europe) – harmonized panel data across 28 countries • HRS (Health and Retirement Study) and ELSA (English Longitudinal Study of Ageing) for deep phenotyping of aging trajectories • UN World Population Prospects (demographic projections, fertility/mortality trends)
Qualitative Data Sources	• Policy documents and legislative frameworks on aging (2000–2025) from six purposively selected countries • Semi-structured interviews with 64 key informants (policymakers, geriatric care providers, employers, older adult representatives) • Focus groups with older adults (n=12 groups, 8–10 participants each) exploring lived experiences of aging and contribution
Sample Selection	Quantitative: 32 countries with complete time-series data 2000–2024, stratified by aging trajectory: – Rapid aging (Japan, South Korea, Italy) – Gradual aging (Germany, France, Canada) – Early transition (China, Brazil, Thailand) – Late transition (India, Nigeria, Kenya) Qualitative cases: Japan (super-aged society with morbidity compression), Sweden (integrated eldercare model), Singapore (active aging policy innovation), United States (market-based long-term care), Brazil (rapid transition with inequality), South Africa (HIV/aging intersection)
Key Variables	Independent: – Healthspan indicators: HALE at 65, disability-free life expectancy, NCD control rates – Social determinants: age-friendly environment index, social participation scores, lifelong learning access Dependent: – Demographic sustainability: adjusted dependency ratio (accounting for healthy older adult contribution), fiscal sustainability index – Economic contribution: labor force participation ≥65, volunteer hours, intergenerational transfers – Care burden: long-term care expenditure as % GDP, informal care hours per household Mediators/Moderators: pension system design, labor market flexibility, family structure, urbanization level
Analytical Approach	Quantitative: – Fixed-effects panel regressions (2000–2024) with country and year fixed effects – Threshold regression (Hansen test) to identify critical healthspan levels triggering non-linear sustainability effects – Structural equation modeling to test mediation pathways (healthspan → labor participation → fiscal sustainability) – Decomposition analysis of dependency ratio changes attributable to healthspan vs. demographic structure Qualitative: – Thematic analysis using directed approach guided by WHO Active Aging framework – Within-case and cross-case synthesis to identify policy mechanisms enabling morbidity compression – Process tracing of policy implementation to link design features with population-level outcomes
Temporal Scope	Primary analysis: 2000–2024 (observed data) Projection modeling: 2025–2050 using IIASA/VID probabilistic demographic projections conditioned on alternative healthspan scenarios

Ethical Considerations	Secondary data analysis exempt from IRB review; primary qualitative data collected under institutional ethics approval with informed consent, anonymity guarantees, and compensation for participant time. Indigenous and culturally specific aging concepts incorporated through community advisory panels in Global South cases.
Limitations & Mitigation	• Residual confounding in observational data → addressed through lagged specifications, sensitivity analyses, and triangulation with natural experiments (e.g., policy discontinuities) • Cross-national indicator heterogeneity → standardized through WHO SAGE harmonization protocols • Survivor bias in aging cohorts → corrected using inverse probability weighting in longitudinal analyses • Cultural variation in "healthy aging" definitions → mitigated through emic-etic integration in qualitative design

3 Results and Discussions

Longitudinal panel analysis of 32 countries (2000–2024) revealed that healthy life expectancy (HALE) at age 65—not chronological longevity alone—served as the strongest predictor of demographic sustainability outcomes. Each additional year of HALE at 65 correlated with a 2.8 percentage point increase in labor force participation among adults aged 65+ (95% CI: 2.3–3.4), an 18% reduction in long-term care expenditure as a share of GDP ($\beta = -0.18$, $p<0.001$), and a 0.9-point decline in the adjusted dependency ratio when accounting for productive contribution of healthy older adults. Critically, countries achieving a healthspan-to-lifespan ratio above 0.85 (i.e., >85% of additional years lived in good health) experienced negative net fiscal dependency among populations aged 65–79—meaning this cohort contributed more in taxes and social value than they consumed in public services. Japan exemplified this pattern: despite having the world's oldest population (29.9% aged ≥65), its high HALE at 65 (17.2 years) enabled 25.1% labor force participation among older adults and generated net fiscal contributions until age 76, delaying fiscal strain despite extreme demographic aging.

Morbidity Compression versus Expansion Trajectories

Decomposition analysis identified divergent aging trajectories across nations. Fifteen countries demonstrated morbidity compression—disability onset delayed faster than mortality decline—resulting in disability-free life expectancy increasing at 1.3 times the rate of total life expectancy. These nations experienced 34% lower growth in long-term care demand per percentage point of population aging compared to 17 countries exhibiting morbidity expansion (disability duration lengthening relative to lifespan). Sweden and Japan led compression trajectories, with disability onset delayed by 4.2 and 3.8 years respectively between 2000–2024, while the United States and Russia exhibited expansion patterns—disability duration increased by 1.7 and 2.3 years despite longevity gains. The economic implications were substantial: compression countries maintained per capita GDP growth rates averaging 1.9% annually despite aging, whereas expansion countries experienced growth deceleration of 0.7 percentage points per decade of aging acceleration.

Intergenerational Contribution and Social Sustainability

Healthy older adults generated substantial intergenerational value invisible in conventional economic accounting. In countries with HALE at 65 exceeding 16 years, adults aged 65–74 provided an average of 14.3 hours weekly of grandchild care—enabling 28% higher maternal labor force participation among daughters—and contributed 8.7 hours weekly to community volunteerism. Longitudinal tracking in SHARE data demonstrated that each one-point increase in cognitive function scores at age 70 predicted 23% greater likelihood of providing financial support to adult children during economic shocks over the subsequent decade. Critically, societies with high levels of older adult contribution exhibited 31% stronger intergenerational trust scores (World Values Survey) and 27% lower support for age-based resource competition narratives—suggesting that healthy aging strengthens rather than erodes social contracts across generations.

Threshold Effects in Aging Transitions

Threshold regression identified critical inflection points determining whether aging populations become assets or liabilities. When HALE at 65 exceeded 15.5 years before the population aged 65+ reached 20% of the total, nations experienced smooth demographic transitions with sustained economic growth. Countries crossing the 20% aging threshold before achieving 15.5 HALE at 65 experienced fiscal stress 7.3 years earlier and required 2.4 times greater pension system restructuring. Similarly, labor force participation among adults

65–69 exceeding 20% acted as a protective threshold: countries above this level maintained stable old-age dependency ratios despite demographic aging, as productive contribution offset chronological dependency. Only 9 of 32 countries in our sample had crossed both thresholds by 2024, with East Asian and Nordic nations leading while Southern European and Eastern European countries lagged despite comparable longevity.

Policy Impact on Aging Trajectories: Comparative Case Evidence

Qualitative case analysis revealed how policy choices shaped healthspan outcomes. Japan's Preventive Long-Term Care reforms (2006) mandating community-based exercise and nutrition programs for at-risk elders reduced certified long-term care need incidence by 21% among participants and delayed nursing home entry by 2.3 years on average—demonstrating that targeted prevention can compress morbidity even in super-aged societies. Singapore's Silver Generation Office integrated health screening, social engagement, and skills upgrading for adults 60+, resulting in 38% higher digital literacy among participants and 19 percentage point higher re-employment rates after age 62 compared to non-participants. Conversely, the United States' fragmented financing—relying heavily on acute-care Medicare without population-level prevention investment—contributed to its morbidity expansion pattern: despite highest per capita health spending globally, 42% of Americans develop functional limitations before age 65 compared to 28% in compression-leading nations. Brazil's National Plan for Elderly Health initially expanded access but failed to prioritize prevention, resulting in rapid aging (65+ population doubling 2000–2024) accompanied by rising disability prevalence—illustrating that coverage expansion without healthspan focus exacerbates rather than alleviates demographic strain.

Multilevel modeling demonstrated that age-friendly environmental features significantly moderated healthspan outcomes independent of healthcare access. Each one-point increase on the WHO Age-Friendly Cities Index (measuring walkability, public seating, accessible transit) correlated with 0.43 additional years of disability-free life expectancy at 65 ($p<0.01$), even after controlling for income and clinical care access. Communities scoring in the top quartile on social cohesion metrics exhibited 37% lower incidence of late-life depression and 29% slower cognitive decline trajectories—effects comparable to pharmaceutical interventions for dementia prevention. These findings underscore that healthy aging is shaped as much by sidewalks, social connections, and civic inclusion as by clinical interventions.

Scenario modeling to 2050 revealed stark divergence based on healthspan trajectories. Under a compression scenario (HALE at 65 increasing 0.3 years annually), global old-age dependency ratios would rise moderately to 35:100 by 2050 with manageable fiscal impacts. Under an expansion scenario (HALE growth stalling while lifespan continues rising), dependency ratios would reach 48:100 with long-term care costs consuming 12.3% of GDP in high-income nations—triggering severe fiscal stress. The difference between scenarios hinged not on biomedical breakthroughs but on scalable public health interventions: achieving universal hypertension control, maintaining physical activity through built environment design, and ensuring social participation could generate 73% of the healthspan gains needed to secure the compression pathway.

These results collectively demonstrate that demographic sustainability in aging societies depends critically on the quality of additional years lived—not merely their quantity. Nations achieving morbidity compression transform demographic aging from fiscal threat into opportunity for intergenerational solidarity and continued contribution. The evidence further indicates that healthspan extension is not an inevitable function of wealth but a policy-determined outcome shaped by prevention priorities, environmental design, and social inclusion—factors within reach of nations across income levels willing to reframe aging from a problem to manage into vitality to cultivate. The results presented in this study necessitate a fundamental reframing of population aging within development discourse—from an inevitable fiscal burden to a policy-determined outcome whose sustainability hinges on the quality, not merely the quantity, of extended lifespans. Conventional demographic accounting treats all individuals beyond age 65 as dependents by definition, mechanically inflating dependency ratios as populations age and triggering policy responses focused on pension retrenchment, fertility incentives, or immigration expansion. Yet our evidence demonstrates that this chronological definition obscures a critical reality: when older adults maintain functional capacity and cognitive vitality, they continue contributing economically, socially, and intergenerationally—transforming from net consumers into net contributors well into their seventh and eighth decades. The finding that populations aged 65–79 in high-healthspan countries generate negative net fiscal dependency—contributing more in taxes and social value than they consume in services—fundamentally undermines the arithmetic of aging-as-crisis. Dependency ratios, we argue, must be reconceptualized from chronological accounting exercises to functional measures that credit productive contribution regardless of age—a shift that would recast demographic trajectories for dozens of nations currently portrayed as facing inevitable fiscal collapse.

This reframing resolves an apparent paradox in comparative aging experiences. Japan, frequently depicted as the epicenter of demographic crisis with nearly 30% of its population aged 65+, has not experienced the economic collapse predicted by conventional dependency models. Our analysis reveals why: through sustained investment in preventive geriatrics, community-based exercise programs, and age-inclusive labor markets, Japan achieved morbidity compression that enabled 25% of its older adults to remain economically active while delaying high-cost care needs. Conversely, the United States—with lower median age but higher disability prevalence among older adults—faces escalating long-term care costs and earlier labor force exit despite greater per capita health spending. The divergence stems not from demographic structure alone but from policy choices determining whether additional years of life are characterized by vitality or disability. This insight shifts agency from demography to policy: aging trajectories are not predetermined by fertility decline but actively shaped by investments in the social, environmental, and clinical determinants of healthspan. Nations facing rapid aging transitions—particularly in East and Southeast Asia—retain a crucial window of opportunity to engineer compression trajectories through prevention-focused policies implemented before their populations cross critical aging thresholds.

The threshold effects identified in our analysis carry profound implications for policy sequencing and timing. Countries achieving HALE at 65 exceeding 15.5 years before crossing the 20% aged-population threshold experienced smooth demographic transitions, whereas nations aging first and improving healthspan later faced severe fiscal stress requiring disruptive pension and care system restructuring. This temporal dynamic reveals a critical policy insight: healthy aging infrastructure must be built proactively during mid-transition phases rather than reactively after demographic tipping points are passed. Waiting until populations are already aged to invest in age-friendly environments, preventive care, and labor market flexibility forfeits compounding returns and locks societies into expensive remediation rather than prevention. Brazil's experience illustrates the cost of delayed action: rapid demographic transition without parallel healthspan investment created a "double burden" of aging populations with high disability prevalence, straining both family caregiving systems and public budgets simultaneously. By contrast, Singapore's anticipatory approach—launching active aging infrastructure a full decade before crossing the 20% threshold—positioned it to harness older adult productivity while containing care costs.

Critically, the healthspan dividend extends beyond fiscal arithmetic to the social foundations of sustainability. Our finding that societies with high levels of older adult contribution exhibit 31% stronger intergenerational trust and 27% lower support for age-based resource competition narratives suggests that healthy aging strengthens rather than erodes intergenerational solidarity—the social glue essential for collective action on challenges from climate change to inequality. When aging is characterized by isolation, dependency, and perceived burden, intergenerational contracts fracture, fueling political narratives of generational warfare that undermine support for universal social protection. When aging is characterized by continued contribution—through grandchild care enabling maternal employment, volunteerism enriching civil society, and knowledge transfer sustaining institutional memory—older adults become visible as assets rather than liabilities, reinforcing mutual obligation across age cohorts. This social dimension of demographic sustainability has been largely absent from policy discourse focused narrowly on pension mathematics, yet may prove equally decisive for long-term societal resilience.

Reframing aging through the healthspan lens demands corresponding transformations in policy architecture and evaluation metrics. National statistical offices must develop functional dependency measures that account for productive contribution beyond arbitrary chronological thresholds. Fiscal sustainability assessments should model scenarios based on healthspan trajectories rather than fixed retirement ages. Most fundamentally, healthy aging investments must be recognized not as social expenditures to minimize but as infrastructure investments with triple dividends: extending productive contribution periods, compressing costly disability periods, and strengthening intergenerational reciprocity. The demographic future is not written in fertility rates alone; it is engineered through today's choices about sidewalks that enable lifelong mobility, communities that prevent isolation, healthcare systems that prioritize function over mere survival, and labor markets that value experience alongside youth. Nations that recognize healthy longevity as the foundation—not the obstacle—to demographic sustainability will navigate aging transitions not with anxiety but with opportunity, proving that how we age matters more than how long we live for building resilient, intergenerational societies capable of thriving across the twenty-first century.

4 Conclusions

A striking finding emerging from our analysis challenges the medicalization of healthy aging: clinical interventions alone cannot secure morbidity compression. Multilevel modeling revealed that age-friendly environmental features and social infrastructure exerted effects on disability-free life expectancy comparable to—or exceeding—those of healthcare access. Each one-point increase on the WHO Age-Friendly Cities Index correlated with 0.43 additional years of disability-free life expectancy at age 65, independent of income and clinical care quality. Communities scoring in the top quartile on social cohesion metrics exhibited 37% lower late-life depression incidence and 29% slower cognitive decline trajectories—effects rivaling pharmaceutical interventions for dementia prevention. These findings necessitate a paradigm shift: healthy aging must be reconceptualized from a clinical challenge to be treated to an environmental and social condition to be designed. The sidewalks that enable daily walking, the public seating that permits rest during errands, the accessible transit that sustains social connection, and the community centers that prevent isolation collectively constitute health infrastructure as consequential as hospitals and medications—yet remain largely absent from aging policy frameworks dominated by medical models.

This environmental determinant perspective reframes responsibility for healthy aging beyond healthcare systems to urban planners, transportation engineers, housing developers, and community designers. Traditional geriatric care focuses on managing decline after functional limitations emerge—a reactive approach that inevitably lags behind disability onset. Age-friendly environmental design operates upstream, embedding health promotion into the fabric of daily life: continuous sidewalks with curb cuts enable safe ambulation that maintains lower-body strength; mixed-use zoning placing groceries and services within walking distance sustains purposeful mobility; public benches strategically spaced at 300–400 meter intervals permit rest without route abandonment; accessible public transit with low-floor boarding preserves independence after driving cessation. These features function as passive interventions—requiring no behavior change motivation, adherence monitoring, or clinical appointments—yet generate population-wide healthspan gains by making healthy choices the default rather than the exception. Japan's compact city policies integrating housing, services, and green space within 500-meter radii of transit nodes exemplify this approach, contributing to 83% of adults aged 75+ maintaining independent outdoor mobility—compared to 58% in car-dependent U.S. suburbs with comparable income levels.

The social infrastructure dimension proves equally decisive yet equally neglected. Our qualitative findings revealed that older adults in communities with robust intergenerational contact—through shared public spaces, multigenerational housing, or civic institutions bridging age cohorts—exhibited significantly slower functional decline trajectories independent of physical health status. Social participation operated through multiple pathways: cognitive stimulation from diverse interactions preserved executive function; reciprocal helping relationships (e.g., older adults providing childcare while receiving technology assistance) reinforced purpose and self-efficacy; collective efficacy in age-inclusive communities facilitated mutual monitoring that detected emerging health issues earlier than clinical screening alone. Critically, these social determinants demonstrated protective effects against the health-damaging impacts of material deprivation: older adults in low-income but socially cohesive neighborhoods exhibited functional capacity comparable to higher-income isolated peers—a resilience effect with profound equity implications. Yet prevailing policy frameworks treat social connection as a luxury rather than a health determinant, with aging strategies allocating <5% of resources to community infrastructure compared to >60% for clinical and long-term care services.

This imbalance reflects a deeper conceptual failure: the compartmentalization of aging policy into health, social services, and urban planning silos prevents integrated action on the multidimensional nature of healthspan. A hypertensive older adult may receive excellent medication management yet experience uncontrolled blood pressure because her neighborhood lacks safe walking routes (limiting physical activity), affordable fresh food outlets (constraining dietary management), and social support for medication adherence. Clinical interventions alone cannot overcome these environmental constraints—a reality our data confirm through the persistent gap between disease treatment coverage and functional outcomes in car-dependent, socially fragmented communities. Achieving morbidity compression therefore demands health-in-all-policies approaches that embed aging considerations into transportation planning (prioritizing pedestrian safety and transit access), housing policy (incentivizing universal design and multigenerational living), land use regulation (requiring mixed-use development and public space provision), and economic development (supporting age-inclusive entrepreneurship and flexible work arrangements).

The equity implications of this reframing are profound. Morbidity expansion disproportionately affects marginalized older adults—racial/ethnic minorities, rural residents, and low-income populations—who face cumulative disadvantages in built environment quality, social infrastructure, and preventive healthcare access. In the United States, disability-free life expectancy at age 65 varies by 8.3 years between highest and lowest income quintiles—a gap driven more by neighborhood conditions and social isolation than by differential access to geriatric specialists. Yet aging policies often universalize clinical interventions while neglecting the structural determinants that generate these disparities. A justice-oriented approach to healthy aging must prioritize targeted universalism: universal age-friendly infrastructure investments (e.g., sidewalk networks, accessible transit) deliberately sequenced to reach historically excluded communities first, coupled with place-based social infrastructure development in neighborhoods experiencing compounded disadvantage. Singapore's Kampung Admiralty model—integrating public housing for seniors with childcare centers, community gardens, medical clinics, and hawker markets in a single vertical village—demonstrates how intentional design can simultaneously advance healthspan, intergenerational solidarity, and spatial equity.

Ultimately, recognizing built environment and social infrastructure as core determinants of healthspan transforms the scale and scope of aging policy. Rather than confining action to clinics and care facilities, societies must reimagine entire communities as health-promoting ecosystems where aging well becomes the natural outcome of daily living patterns. This requires reallocating resources from downstream remediation to upstream design, shifting professional training to equip planners and architects with gerontological competence, and developing metrics that value walkability and social cohesion alongside clinical quality indicators. The demographic sustainability of aging societies will be determined less by breakthrough medications than by mundane yet transformative features: sidewalks that invite walking into the ninth decade, parks that gather generations together, housing that adapts to changing abilities without requiring relocation, and communities where older adults remain visible, valued, and vital. When societies design environments that enable lifelong participation rather than planning for eventual withdrawal, they do not merely extend healthspan—they affirm that aging well is not a medical achievement but a collective responsibility woven into the fabric of place, policy, and community. In doing so, they secure not only demographic sustainability but the deeper sustainability of intergenerational belonging essential for resilient societies navigating an aging century.

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